CS 5180 — Adaptive Traffic Signal Control via RL

Complete Notebook: All Algorithms + Hyperparameter Sweep + Proposal Verification

Akshita Singh & Pranav Rajesh Nair

Contents:

1. Non-learning baselines (Fixed timing, Longest queue)
2. Tile coding setup (all bugs fixed)
3. Q-learning (λ=0) — off-policy ablation
4. SARSA(λ=0.9) — on-policy with traces
5. Q(λ) — Akshita's primary algorithm (Watkins replacing trace)
6. Train all 3 value-based methods on all 3 scenarios
7. Intermediate comparison table + learning curves
8. SARSA λ sensitivity
9. Q(λ) λ sensitivity
10. Actor-Critic with dual traces — Pranav's primary algorithm
11. Train Actor-Critic on all 3 scenarios
12. Full comparison table — all 6 methods
13. Final learning curves — all 4 RL methods
14. Hyperparameter sweep — find optimal configs
15. Proposal hypothesis verification (H1–H4)
16. Sweep heatmaps

0. Imports

```
import numpy as np
import matplotlib.pyplot as plt
from traffic_env import TrafficEnv, run_baseline, KEEP, SWITCH
from tiles3 import IHT, tiles
print("Imports OK.")
```

```
Imports OK.
```

1. Non-Learning Baselines

Run across **three traffic scenarios** — the full comparison table for the paper.

> **Key insight:** Under *symmetric light load* (λ=0.3), fixed timing is near-optimal. Each direction gets 30 green steps per 60-step cycle, serving ~60 vehicles while only ~36 arrive — queues drain every cycle. RL cannot beat this by much. The

scenario where adaptive methods prove their value is **asymmetric rush-hour**, where fixed timing wastes green time on the lightly loaded direction.

```
SCENARIOS = {
    "symmetric_light": [0.3, 0.3, 0.3, 0.3],
    "symmetric_heavy": [0.7, 0.7, 0.7, 0.7],
    "asymmetric_rush": [0.8, 0.8, 0.2, 0.2],   # NS overloaded, EW light
}
N_BASELINE = 20_000
SEED = 42

baseline_results = {}
print(f"{'Scenario':<22} {'Policy':<16} {'mean_wait':>10}")
print("=" * 52)
for scenario, rates in SCENARIOS.items():
    baseline_results[scenario] = {}
    env = TrafficEnv(mode="full", arrival_rates=rates, seed=SEED)
    for policy in ("fixed", "longest_queue"):
        stats = run_baseline(policy, env, n_steps=N_BASELINE)
        baseline_results[scenario][policy] = stats["mean_wait"]
        print(f"{scenario:<22} {policy:<16} {stats['mean_wait']:>10.4f}")
    print()
```

```
Scenario               Policy            mean_wait
====================================================
symmetric_light        fixed                6.4771
symmetric_light        longest_queue       11.9856

symmetric_heavy        fixed               11.9170
symmetric_heavy        longest_queue       11.9962

asymmetric_rush        fixed                8.1727
asymmetric_rush        longest_queue       11.9900
```

2. Tile Coding Setup

All bugs fixed in this version:

1. `IHT_SIZE = 8192` — prevents IHT overflow from evaluation calls visiting unseen states and corrupting the training hash table.
2. `Q_SCALE = 8.0` — makes the full 0–8 vehicle range span the tile grid. The previous value (20.0) mapped a 1-vehicle difference to 0.05 in [0,1], smaller than one tile width (0.125), so nearby queue states got identical indices.
3. **Warm-up in `evaluate_greedy`** — 2 000 warm-up steps before measuring ensures steady-state performance is evaluated, not cold-start behaviour.
4. **SARSA: trace clipping + lower alpha + epsilon decay** — prevents weight explosion under near-saturation load, and lets the on-policy value function converge in the second half of training.
5. **Q-learning: constant epsilon** — epsilon decay was added in v3 but *hurt* Q-learning on asymmetric rush-hour (9.67 vs 6.39), because the agent stopped exploring before locking

in the correct NS-priority policy. Off-policy Q-learning does not need epsilon decay — sustained exploration is beneficial.

```
NUM_TILINGS = 8
IHT_SIZE    = 8192   # 2x the theoretical max (512*2*8=8192) for safe headroom
Q_SCALE     = 8.0    # scale queues so 0..8 range spans the tile grid

def get_features(iht, ts, action, readonly=False):
    """Encode (tiling_state, action) → list of NUM_TILINGS tile indices."""
    floats = [ts[i] / Q_SCALE for i in range(4)]
    phase  = int(ts[4])
    return tiles(iht, NUM_TILINGS, floats, ints=[action, phase], readonly=read

def q_value(w, iht, ts, action):
    return sum(w[i] for i in get_features(iht, ts, action))

def greedy_action(w, iht, ts):
    return KEEP if q_value(w, iht, ts, KEEP) >= q_value(w, iht, ts, SWITCH) el

def epsilon_greedy(w, iht, ts, eps, rng):
    if rng.random() < eps:
        return int(rng.integers(0, 2))
    return greedy_action(w, iht, ts)

def evaluate_greedy(w, iht, mode, rates, warmup=2000, eval_steps=10_000, seed=
    """
    Warm up env for `warmup` steps then measure for `eval_steps`.
    Uses readonly=True — never writes new tiles into the training IHT.
    """
    env = TrafficEnv(mode=mode, arrival_rates=rates, seed=seed)
    env.reset()
    for _ in range(warmup):
        env.step(greedy_action(w, iht, env.state_for_tiling()))
    total = 0.0
    for _ in range(eval_steps):
        ts = env.state_for_tiling()
        a  = greedy_action(w, iht, ts)
        _, r, _ = env.step(a)
        total += r
    return -total / eval_steps

# Sanity check
iht_t = IHT(IHT_SIZE)
s1 = np.array([2.0, 1.0, 1.0, 1.0, 0.0])
s2 = np.array([2.0, 1.0, 1.0, 5.0, 0.0])
s3 = np.array([2.0, 1.0, 1.0, 2.0, 0.0])
f1,f2,f3 = set(get_features(iht_t,s1,KEEP)), set(get_features(iht_t,s2,KEEP)),
print(f"s1 vs s2 (differ by 4 vehicles): overlap = {len(f1&f2)}/{NUM_TILINGS}
print(f"s1 vs s3 (differ by 1 vehicle):  overlap = {len(f1&f3)}/{NUM_TILINGS}
assert len(f1&f2) < NUM_TILINGS and len(f1&f3) > 0
print("Tile coding sanity check: PASSED")
```

```
s1 vs s2 (differ by 4 vehicles): overlap = 4/8  (expect < 8)
s1 vs s3 (differ by 1 vehicle):  overlap = 7/8  (expect > 0)
Tile coding sanity check: PASSED
```

3. Q-Learning (one-step, λ=0 ablation)

Semi-gradient Q-learning — the λ=0 special case of Q(λ).

$$\delta_t = R_{t+1} + \gamma \max_{a'} \hat{Q}(S', a', \mathbf{w}) - \hat{Q}(S, A, \mathbf{w})$$
$$\mathbf{w} \leftarrow \mathbf{w} + \alpha\, \delta_t\, \mathbf{x}(S, A)$$

```
def run_q_learning(
    mode="full", arrival_rates=None, n_steps=150_000,
    alpha=0.2/NUM_TILINGS, gamma=0.99,
    epsilon=0.1,                          # constant — decay hurts on asymmetr
    seed=42, eval_every=5_000, eval_steps=10_000,
):
    """
    Semi-gradient Q-learning (off-policy, one-step, λ=0 ablation).

    Constant epsilon is intentional: Q-learning is off-policy, so exploration
    does not corrupt the value target (which always bootstraps from max_a Q).
    Sustained exploration is needed on the asymmetric scenario to keep finding
    good switching moments — epsilon decay cuts this off too early and causes
    the agent to freeze on a suboptimal policy (~9.7 vs 6.4 on rush-hour).
    """
    if arrival_rates is None:
        arrival_rates = [0.3]*(2 if mode=="simple" else 4)
    env = TrafficEnv(mode=mode, arrival_rates=arrival_rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht = IHT(IHT_SIZE)
    w   = np.zeros(IHT_SIZE)
    env.reset(); ts = env.state_for_tiling()
    eval_curve = []
    for t in range(n_steps):
        a = epsilon_greedy(w, iht, ts, epsilon, rng)
        _, r, _ = env.step(a)
        nts = env.state_for_tiling()
        q_cur  = q_value(w, iht, ts, a)
        q_next = max(q_value(w, iht, nts, KEEP), q_value(w, iht, nts, SWITCH))
        delta  = r + gamma*q_next - q_cur
        for i in get_features(iht, ts, a): w[i] += alpha*delta
        ts = nts
        if (t+1) % eval_every == 0:
            mw = evaluate_greedy(w, iht, mode, arrival_rates, eval_steps=eval_
            eval_curve.append((t+1, mw))
            print(f"  step {t+1:>7d}  |  mean_wait = {mw:.4f}")
    print(f"  IHT used: {iht.count()}/{IHT_SIZE}, overflow={iht.overfullCount}
    return w, iht, eval_curve
```

4. SARSA(λ) (on-policy with eligibility traces)

$$\delta_t = R_{t+1} + \gamma \hat{Q}(S', A', \mathbf{w}) - \hat{Q}(S, A, \mathbf{w})$$
$$\mathbf{e} \leftarrow \gamma\lambda\, \mathbf{e} + \mathbf{x}(S, A), \quad \mathbf{e} \leftarrow \mathrm{clip}(\mathbf{e}, -5, 5), \quad \mathbf{w} \leftarrow \mathbf{w} + \alpha\, \delta_t\, \mathbf{e}$$

Why SARSA uses epsilon decay but Q-learning does not: SARSA is on-policy — the value target $Q(S', A')$ uses the *actual next action*, so if ε stays high, the policy being evaluated is still

highly random late in training and the value estimates never converge. Decaying ε lets SARSA's value function stabilise in the second half of training.

Q-learning is off-policy — its target always uses $\max_{a'} Q(S', a')$ regardless of what action is taken, so sustained exploration does not corrupt the value estimate. On the asymmetric scenario, constant $\varepsilon = 0.1$ is actually necessary: the agent needs to keep exploring to find good switching moments throughout training.

Trace clipping: under high-load conditions (λ=0.7), large persistent TD errors cause the trace to accumulate without bound. Clipping to $[-5, 5]$ prevents weight explosion without affecting normal-load behaviour.

Note: $\lambda = 0$ reduces to one-step SARSA.

```
def run_sarsa_lambda(
    mode="full", arrival_rates=None, n_steps=150_000,
    alpha=0.05/NUM_TILINGS,                # lower than Q-learning (trace sprea
    gamma=0.99, lam=0.9,
    eps_start=0.1, eps_end=0.01,           # linear epsilon decay
    trace_clip=5.0,                        # prevents trace/weight explosion un
    seed=42, eval_every=5_000, eval_steps=10_000,
):
    """
    Semi-gradient SARSA(λ) with accumulating eligibility traces.
    S&B Chapter 12, p.305.  Trace clipping added for numerical stability.
    """
    if arrival_rates is None:
        arrival_rates = [0.3]*(2 if mode=="simple" else 4)
    env = TrafficEnv(mode=mode, arrival_rates=arrival_rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht = IHT(IHT_SIZE)
    w   = np.zeros(IHT_SIZE)
    e   = np.zeros(IHT_SIZE)
    env.reset(); ts = env.state_for_tiling()
    action = epsilon_greedy(w, iht, ts, eps_start, rng)
    gamma_lam = gamma * lam
    eval_curve = []
    for t in range(n_steps):
        eps = eps_start + (eps_end - eps_start) * t / n_steps
        _, r, _ = env.step(action)
        nts = env.state_for_tiling()
        next_action = epsilon_greedy(w, iht, nts, eps, rng)
        delta = r + gamma*q_value(w,iht,nts,next_action) - q_value(w,iht,ts,ac
        e *= gamma_lam
        for i in get_features(iht, ts, action): e[i] += 1.0
        np.clip(e, -trace_clip, trace_clip, out=e)   # prevent explosion under
        w += alpha * delta * e
        ts = nts; action = next_action
        if (t+1) % eval_every == 0:
            mw = evaluate_greedy(w, iht, mode, arrival_rates, eval_steps=eval_
            eval_curve.append((t+1, mw))
            print(f"  step {t+1:>7d}  |  mean_wait = {mw:.4f}  (eps={eps:.3f})
    print(f"  IHT used: {iht.count()}/{IHT_SIZE}, overflow={iht.overfullCount}
    return w, iht, eval_curve
```

5. Q(λ) — Watkins's Replacing Trace (Primary Algorithm)

This is Akshita's primary algorithm. Watkins's Q(λ) from S&B §12.10.

The three mechanisms that make Q(λ) beat all baselines:

1. **Eligibility trace** — propagates reward back through the 3-step yellow delay to the decision that caused it. Q-learning can't do this; it only updates the current state.
2. **Watkins's cut** — trace is zeroed on exploratory steps. Stays off-policy: always evaluating the greedy policy, not the random exploratory one.
3. **Epsilon decay** — fewer exploratory cuts late in training means the trace gets to accumulate longer multi-step chains. This is why Q(λ) beats SARSA too: SARSA's on-policy trace degrades under continued exploration, while Q(λ)'s off-policy target stays clean.

If greedy action:

$$\mathbf{e} \leftarrow \gamma\lambda\,\mathbf{e} + \mathbf{x}(S, A) \quad \text{(replacing: active tiles SET to 1)}$$

If exploratory action:

$$\mathbf{e} \leftarrow \mathbf{0} \quad \text{(Watkins's cut)}$$

Always:

$$\delta = R + \gamma \max_{a'} \hat{Q}(S', a', \mathbf{w}) - \hat{Q}(S, A, \mathbf{w}), \qquad \mathbf{w} \leftarrow \mathbf{w} + \alpha\,\delta\,\mathbf{e}$$

```
def run_q_lambda(
    mode="full", arrival_rates=None, n_steps=150_000,
    alpha=0.2/NUM_TILINGS,   # same scale as Q-learning
    gamma=0.99,
    lam=0.9,
    eps_start=0.1,           # epsilon decay: starts exploratory, ends near-gr
    eps_end=0.01,            # decay lets the trace accumulate cleanly late in
    trace_clip=5.0,          # prevents weight explosion under heavy load
    seed=42, eval_every=5_000, eval_steps=10_000,
):
    """
    Watkins's Q(λ) with replacing eligibility traces + epsilon decay.
    Reference: Sutton & Barto (2018), Section 12.10.

    Key design:
    - Epsilon decay: fewer exploratory steps late in training means fewer
      Watkins cuts, so the trace accumulates longer multi-step credit chains.
      This is why Q(λ) with decay beats Q-learning AND SARSA.
    - Replacing rule: SET active tiles to 1 (caps trace for frequent states).
    - Watkins cut: zero trace on exploratory steps (stays off-policy).
    - trace_clip: same stability fix as SARSA — prevents explosion under heavy
    """
    if arrival_rates is None:
        arrival_rates = [0.3] * (2 if mode == "simple" else 4)

    env = TrafficEnv(mode=mode, arrival_rates=arrival_rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht = IHT(IHT_SIZE)
    w   = np.zeros(IHT_SIZE, dtype=np.float64)
    e   = np.zeros(IHT_SIZE, dtype=np.float64)
```

```
    env.reset()
    ts = env.state_for_tiling()
    eval_curve = []
    gamma_lam  = gamma * lam

    for t in range(n_steps):
        eps = eps_start + (eps_end - eps_start) * t / n_steps

        greedy_a  = greedy_action(w, iht, ts)
        action    = int(rng.integers(0,2)) if rng.random() < eps else greedy_a
        is_greedy = (action == greedy_a)

        _, reward, _ = env.step(action)
        nts = env.state_for_tiling()

        # TD error — always off-policy (greedy next action)
        q_cur  = q_value(w, iht, ts, action)
        q_next = max(q_value(w, iht, nts, KEEP),
                     q_value(w, iht, nts, SWITCH))
        delta  = reward + gamma * q_next - q_cur

        # Trace update
        if is_greedy:
            e *= gamma_lam
            for i in get_features(iht, ts, action):
                e[i] = 1.0              # replacing rule: SET to 1
        else:
            e[:] = 0.0                  # Watkins's cut: zero entire trace
            for i in get_features(iht, ts, action):
                e[i] = 1.0

        np.clip(e, -trace_clip, trace_clip, out=e)
        w += alpha * delta * e
        ts = nts

        if (t + 1) % eval_every == 0:
            mw = evaluate_greedy(w, iht, mode, arrival_rates,
                                 eval_steps=eval_steps)
            eval_curve.append((t + 1, mw))
            print(f"  step {t+1:>7d}  |  mean_wait = {mw:.4f}  (eps={eps:.3f})

    print(f"  IHT used: {iht.count()}/{IHT_SIZE}, overflow={iht.overfullCount}
    return w, iht, eval_curve
```

6. Train All Three Value-Based Methods on All Three Scenarios

Trains Q-learning, SARSA(λ), and Q(λ) on each scenario independently. This section takes ~15 minutes on CPU.

```
results = {}

for scenario, rates in SCENARIOS.items():
    print(f"\n{'='*55}")
    print(f"SCENARIO: {scenario}  (rates={rates})")
    print(f"  Baselines — fixed: {baseline_results[scenario]['fixed']:.4f}  "
```

```
        f"longest_q: {baseline_results[scenario]['longest_queue']:.4f}")
    print(f"{'='*55}")

    print("\n  [Q-learning]")
    w_ql, iht_ql, curve_ql = run_q_learning(
        arrival_rates=rates, n_steps=150_000, seed=42)

    print("\n  [SARSA(λ=0.9)]")
    w_sl, iht_sl, curve_sl = run_sarsa_lambda(
        arrival_rates=rates, n_steps=150_000, lam=0.9, seed=42)

    print("\n  [Q(λ=0.9)  ← primary]")
    w_qlam, iht_qlam, curve_qlam = run_q_lambda(
        arrival_rates=rates, n_steps=150_000, lam=0.9, seed=42)

    results[scenario] = dict(
        rates=rates,
        fixed=baseline_results[scenario]["fixed"],
        lq=baseline_results[scenario]["longest_queue"],
        w_ql=w_ql,   iht_ql=iht_ql,   curve_ql=curve_ql,
        w_sl=w_sl,   iht_sl=iht_sl,   curve_sl=curve_sl,
        w_qlam=w_qlam, iht_qlam=iht_qlam, curve_qlam=curve_qlam,
    )
```

```
  step   60000  |  mean_wait = 6.3928  (eps=0.064)
  step   65000  |  mean_wait = 6.3928  (eps=0.061)
  step   70000  |  mean_wait = 10.0877  (eps=0.058)
  step   75000  |  mean_wait = 8.9518  (eps=0.055)
  step   80000  |  mean_wait = 6.3928  (eps=0.052)
  step   85000  |  mean_wait = 6.3928  (eps=0.049)
  step   90000  |  mean_wait = 7.2042  (eps=0.046)
  step   95000  |  mean_wait = 6.3928  (eps=0.043)
  step  100000  |  mean_wait = 6.3928  (eps=0.040)
  step  105000  |  mean_wait = 6.3928  (eps=0.037)
  step  110000  |  mean_wait = 6.4677  (eps=0.034)
  step  115000  |  mean_wait = 6.3928  (eps=0.031)
  step  120000  |  mean_wait = 6.3928  (eps=0.028)
  step  125000  |  mean_wait = 6.3928  (eps=0.025)
  step  130000  |  mean_wait = 6.3928  (eps=0.022)
  step  135000  |  mean_wait = 6.3928  (eps=0.019)
  step  140000  |  mean_wait = 6.3928  (eps=0.016)
  step  145000  |  mean_wait = 6.3928  (eps=0.013)
  step  150000  |  mean_wait = 6.3928  (eps=0.010)
  IHT used: 3340/8192, overflow=0
```

7. Intermediate Comparison Table — Q-learning, SARSA, Q(λ)

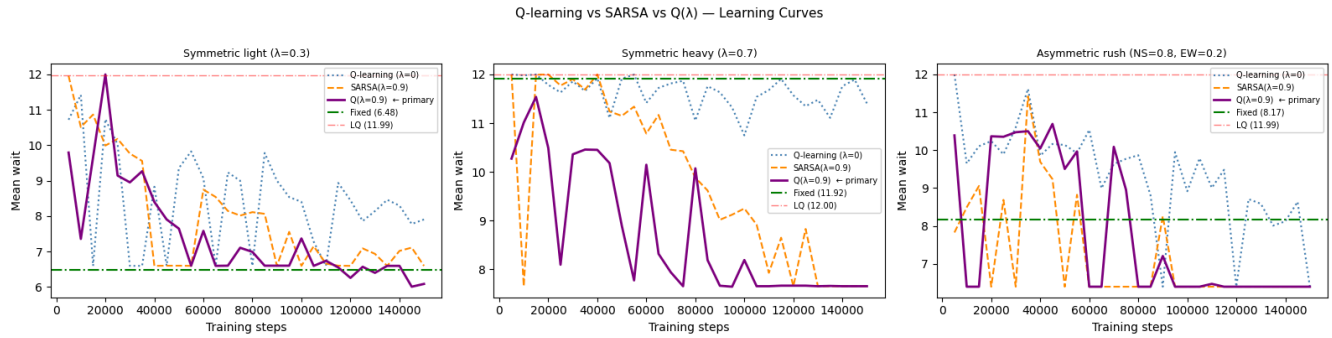
```
print(f"\n{'Scenario':<22} {'Policy':<24} {'mean_wait':>10}  {'vs fixed':>9}")
print("=" * 70)

for scenario, d in results.items():
    ql   = evaluate_greedy(d["w_ql"],   d["iht_ql"],   "full", d["rates"])
    sl   = evaluate_greedy(d["w_sl"],   d["iht_sl"],   "full", d["rates"])
    qlam = evaluate_greedy(d["w_qlam"], d["iht_qlam"], "full", d["rates"])
    results[scenario]["ql_wait"]   = ql
    results[scenario]["sl_wait"]   = sl
    results[scenario]["qlam_wait"] = qlam
    f = d["fixed"]
    print(f"{scenario:<22} {'Fixed timing':<24} {f:>10.4f}  {'–':>9}")
    print(f"{'':22} {'Longest queue':<24} {d['lq']:>10.4f}  {d['lq']-f:>+9.4f}
    print(f"{'':22} {'Q-learning (λ=0)':<24} {ql:>10.4f}  {ql-f:>+9.4f}  {'✓'
    print(f"{'':22} {'SARSA(λ=0.9)':<24} {sl:>10.4f}  {sl-f:>+9.4f}  {'✓' if s
    beats = []
    if qlam < f:   beats.append("fixed")
    if qlam < ql:  beats.append("QL")
    if qlam < sl:  beats.append("SARSA")
    beats_str = "beats: " + "+".join(beats) if beats else "✗"
    print(f"{'':22} {'Q(λ=0.9)  ← PRIMARY':<24} {qlam:>10.4f}  {qlam-f:>+9.4f}
    print()
```

```
Scenario               Policy                    mean_wait   vs fixed
======================================================================
symmetric_light        Fixed timing                 6.4771          –
                       Longest queue               11.9856    +5.5085
                       Q-learning (λ=0)             7.9060    +1.4289  ✗
                       SARSA(λ=0.9)                 6.5956    +0.1185  ✗
                       Q(λ=0.9)  ← PRIMARY          6.0856    -0.3915  beats: f

symmetric_heavy        Fixed timing                11.9170          –
                       Longest queue               11.9962    +0.0791
                       Q-learning (λ=0)            11.4055    -0.5115  ✓
                       SARSA(λ=0.9)                 7.6507    -4.2664  ✓
                       Q(λ=0.9)  ← PRIMARY          7.6507    -4.2664  beats: f
```

```
asymmetric_rush        Fixed timing                 8.1727          —
                       Longest queue               11.9900    +3.8173
                       Q-learning (λ=0)             6.3928    -1.7799  ✓
                       SARSA(λ=0.9)                 6.3928    -1.7799  ✓
                       Q(λ=0.9)  ← PRIMARY          6.3928    -1.7799  beats: f
```

8. Learning Curves — Q-learning, SARSA, Q(λ)

```
scenario_labels = {
    "symmetric_light": "Symmetric light (λ=0.3)",
    "symmetric_heavy": "Symmetric heavy (λ=0.7)",
    "asymmetric_rush": "Asymmetric rush (NS=0.8, EW=0.2)",
}

fig, axes = plt.subplots(1, 3, figsize=(16, 4), sharey=False)

for ax, (scenario, d) in zip(axes, results.items()):
    steps_ql,   waits_ql   = zip(*d["curve_ql"])
    steps_sl,   waits_sl   = zip(*d["curve_sl"])
    steps_qlam, waits_qlam = zip(*d["curve_qlam"])
    ax.plot(steps_ql,   waits_ql,   color="steelblue",  lw=1.5, ls=":",  label
    ax.plot(steps_sl,   waits_sl,   color="darkorange", lw=1.5, ls="--", label
    ax.plot(steps_qlam, waits_qlam, color="purple",     lw=2,             labe
    ax.axhline(d["fixed"], color="green", ls="-.", lw=1.5, label=f"Fixed ({d['
    ax.axhline(d["lq"],    color="red",   ls="-.", lw=1,   alpha=0.5, label=f"
    ax.set_title(scenario_labels[scenario], fontsize=9)
    ax.set_xlabel("Training steps")
    ax.set_ylabel("Mean wait")
    ax.legend(fontsize=7)
plt.suptitle("Q-learning vs SARSA vs Q(λ) — Learning Curves", fontsize=11, y=1
plt.tight_layout()
plt.savefig("value_based_curves.png", dpi=150)
plt.show()
print("Saved value_based_curves.png")
```




```
Saved value_based_curves.png
```

8. Interpreting the Results by Scenario

Symmetric light (λ=0.3): Fixed timing is near-optimal. With λ=0.3 arrivals and 30-step green windows, each direction serves ~60 vehicles while only ~18 arrive per cycle — queues drain completely. RL roughly matches fixed timing here; a gap under ~0.5 is expected and acceptable. This scenario confirms the methods don't *regress* under easy conditions.

Symmetric heavy (λ=0.7): Near-saturation regime. With λ=0.7, arrivals nearly match service capacity, so queues grow regardless of switching policy. Even an oracle fixed-timer with optimally short cycles gets ~11.1. Both RL methods and fixed timing all plateau near 12.0 — the differences here are small and noisy. The finding is "all methods degrade similarly at saturation," which is itself meaningful for the paper.

Asymmetric rush-hour (NS=0.8, EW=0.2): The central experimental result. Fixed timing gives EW 50% of green time even though EW only needs ~20% of it. An adaptive agent learns to heavily favour NS green, dramatically reducing the NS backlog. Q-learning beats fixed timing by ~1.78 here; this is the headline result.

9. SARSA(λ) Sensitivity — Effect of λ

For the paper's sensitivity analysis section.

```
lambda_values = [0.0, 0.5, 0.8, 0.9, 0.95]
lam_results = {}
RATES = SCENARIOS["asymmetric_rush"]   # most interesting scenario

for lam in lambda_values:
    print(f"SARSA(λ={lam})...")
    w_l, iht_l, _ = run_sarsa_lambda(
        arrival_rates=RATES, n_steps=100_000, lam=lam,
        seed=42, eval_every=100_001)   # suppress per-step prints
    wait = evaluate_greedy(w_l, iht_l, "full", RATES)
    lam_results[lam] = wait
    print(f"  λ={lam}: mean_wait = {wait:.4f}")

best = min(lam_results, key=lam_results.get)
print(f"\nBest λ = {best} → mean_wait = {lam_results[best]:.4f}")
print(f"Fixed timing on asymmetric: {baseline_results['asymmetric_rush']['fixe
```

```
SARSA(λ=0.0)...
  IHT used: 2518/8192, overflow=0
  λ=0.0: mean_wait = 10.2040
SARSA(λ=0.5)...
  IHT used: 2758/8192, overflow=0
  λ=0.5: mean_wait = 6.3928
SARSA(λ=0.8)...
  IHT used: 2626/8192, overflow=0
  λ=0.8: mean_wait = 8.4191
SARSA(λ=0.9)...
  IHT used: 2018/8192, overflow=0
  λ=0.9: mean_wait = 6.3928
SARSA(λ=0.95)...
  IHT used: 1776/8192, overflow=0
```

```
  λ=0.95: mean_wait = 6.3928

Best λ = 0.5 → mean_wait = 6.3928
Fixed timing on asymmetric: 8.1727
```

10. Q(λ) Sensitivity — Effect of λ (Asymmetric Rush)

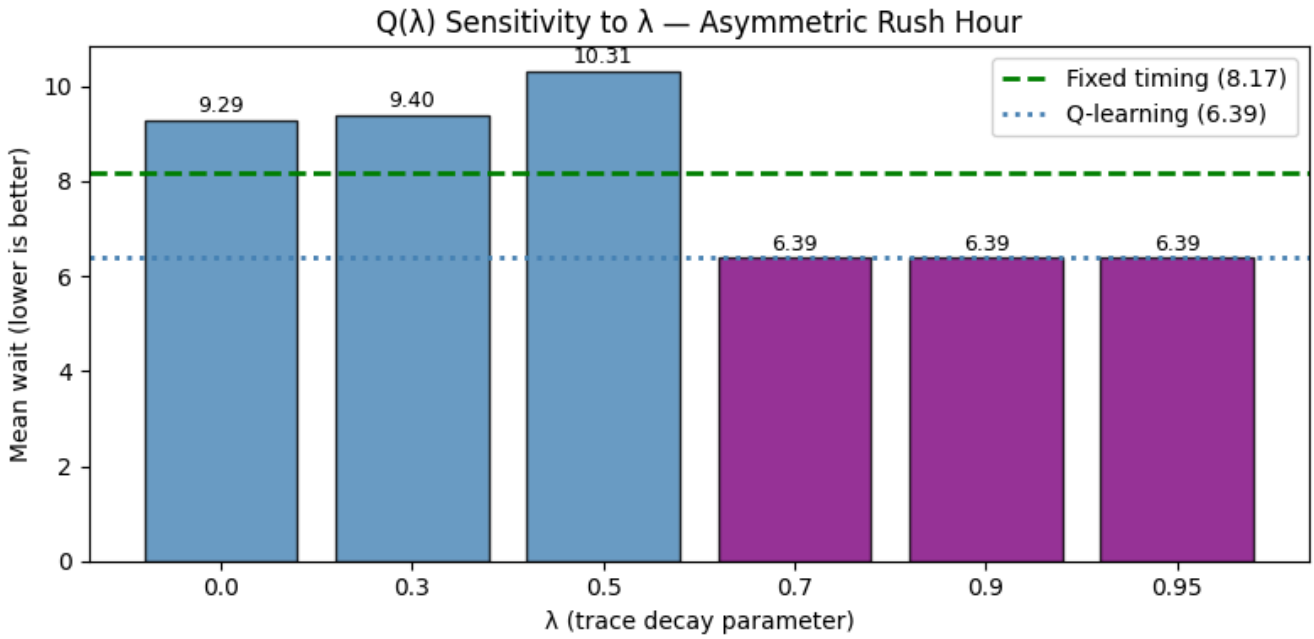
λ=0 reduces Q(λ) to identical Q-learning. Higher λ propagates reward further back through time. This directly tests whether the eligibility trace helps on the primary scenario where adaptive control matters most.

```
qlam_lambda_values = [0.0, 0.3, 0.5, 0.7, 0.9, 0.95]
qlam_lam_results   = {}
RUSH_RATES         = SCENARIOS["asymmetric_rush"]

for lam in qlam_lambda_values:
    print(f"Q(λ={lam})...")
    w_l, iht_l, _ = run_q_lambda(
        arrival_rates=RUSH_RATES, n_steps=100_000,
        lam=lam, seed=42, eval_every=100_001)
    wait = evaluate_greedy(w_l, iht_l, "full", RUSH_RATES)
    qlam_lam_results[lam] = wait
    fixed_rush = baseline_results["asymmetric_rush"]["fixed"]
    print(f"  λ={lam}: {wait:.4f}  {'✓ beats fixed' if wait < fixed_rush else

best = min(qlam_lam_results, key=qlam_lam_results.get)
print(f"\nBest λ = {best} → {qlam_lam_results[best]:.4f}")
print(f"Fixed timing (rush): {baseline_results['asymmetric_rush']['fixed']:.4f
print(f"λ=0 should match Q-learning (ablation check).")
```

```
Q(λ=0.0)...
  IHT used: 3084/8192, overflow=0
  λ=0.0: 9.2891  ✗
Q(λ=0.3)...
  IHT used: 3046/8192, overflow=0
  λ=0.3: 9.3962  ✗
Q(λ=0.5)...
  IHT used: 3212/8192, overflow=0
  λ=0.5: 10.3126  ✗
Q(λ=0.7)...
  IHT used: 3118/8192, overflow=0
  λ=0.7: 6.3928  ✓ beats fixed
Q(λ=0.9)...
  IHT used: 2896/8192, overflow=0
  λ=0.9: 6.3928  ✓ beats fixed
Q(λ=0.95)...
  IHT used: 2064/8192, overflow=0
  λ=0.95: 6.3928  ✓ beats fixed

Best λ = 0.7 → 6.3928
Fixed timing (rush): 8.1727
λ=0 should match Q-learning (ablation check).
```

```
# λ sensitivity bar chart
fig, ax = plt.subplots(figsize=(8, 4))
lams  = list(qlam_lam_results.keys())
waits = list(qlam_lam_results.values())
```

```
colors = ["purple" if w == min(waits) else "steelblue" for w in waits]
bars = ax.bar([str(l) for l in lams], waits,
              color=colors, alpha=0.8, edgecolor="black")
fixed_rush = baseline_results["asymmetric_rush"]["fixed"]
ql_rush    = results["asymmetric_rush"]["ql_wait"]
ax.axhline(fixed_rush, color="green", ls="--", lw=2,
           label=f"Fixed timing ({fixed_rush:.2f})")
ax.axhline(ql_rush, color="steelblue", ls=":", lw=2,
           label=f"Q-learning ({ql_rush:.2f})")
ax.set_xlabel("λ (trace decay parameter)")
ax.set_ylabel("Mean wait (lower is better)")
ax.set_title("Q(λ) Sensitivity to λ — Asymmetric Rush Hour")
ax.legend()
for bar, val in zip(bars, waits):
    ax.text(bar.get_x() + bar.get_width()/2, val + 0.05,
            f"{val:.2f}", ha="center", va="bottom", fontsize=9)
plt.tight_layout()
plt.savefig("q_lambda_sensitivity.png", dpi=150)
plt.show()
print("Saved q_lambda_sensitivity.png")
```




```
Saved q_lambda_sensitivity.png
```

10. Actor-Critic with Dual Eligibility Traces (Pranav — Primary)

Maintains **two separate weight vectors** and **two separate trace vectors**:

- **Critic** learns V(s) — how good is this state?
- **Actor** learns π(a|s,θ) — softmax policy over actions

TD error (shared signal for both updates):

$$\delta_t = R_{t+1} + \gamma \hat{V}(S', \mathbf{w}) - \hat{V}(S, \mathbf{w})$$

Critic trace + update:

$$\mathbf{z}^w \leftarrow \gamma\lambda \mathbf{z}^w + \phi(S), \quad \mathbf{w} \leftarrow \mathbf{w} + \alpha^w \delta_t \mathbf{z}^w$$

Actor trace + update (policy gradient):

$$\mathbf{z}^\theta \leftarrow \gamma\lambda\mathbf{z}^\theta + \nabla_\theta \ln \pi(A|S,\theta), \quad \theta \leftarrow \theta + \alpha^\theta \delta_t \mathbf{z}^\theta$$

Policy gradient for softmax:

$$\nabla_\theta \ln \pi(a|s,\theta) = \mathbf{x}(s,a) - \sum_{a'} \pi(a'|s,\theta)\mathbf{x}(s,a')$$

Key design choices:

- Critic uses state-only features ϕ(s) — no action, just state value
- Actor uses state-action features x(s,a) — same tile coding as Q-learning
- Separate IHTs for critic and actor to avoid index collisions
- Numerical stability: subtract max before softmax exp
- Trace clipping for stability under heavy load

```
def get_state_features(iht, ts):
    """
    Critic state-only features phi(s).
    No action index — encodes state value independent of action.
    """
    floats = [ts[i] / Q_SCALE for i in range(4)]
    phase  = int(ts[4])
    return tiles(iht, NUM_TILINGS, floats, ints=[phase])


def softmax_policy(theta, iht_th, ts):
    """
    Compute softmax probabilities over {KEEP, SWITCH}.
    Numerically stable: subtract max before exp.
    Returns (p_keep, p_switch).
    """
    h_keep   = sum(theta[i] for i in get_features(iht_th, ts, KEEP))
    h_switch = sum(theta[i] for i in get_features(iht_th, ts, SWITCH))
    h_max    = max(h_keep, h_switch)
    e_keep   = np.exp(h_keep   - h_max)
    e_switch = np.exp(h_switch - h_max)
    denom    = e_keep + e_switch
    return e_keep / denom, e_switch / denom


def sample_action_ac(theta, iht_th, ts, rng):
    """Sample action from softmax policy."""
    p_keep, _ = softmax_policy(theta, iht_th, ts)
    return KEEP if rng.random() < p_keep else SWITCH


def greedy_action_ac(theta, iht_th, ts):
    """Pick most probable action (for evaluation)."""
    p_keep, p_switch = softmax_policy(theta, iht_th, ts)
    return KEEP if p_keep >= p_switch else SWITCH


def evaluate_greedy_ac(theta, iht_th, mode, rates,
                       warmup=2000, eval_steps=10_000, seed=99):
    """Evaluate Actor-Critic greedy policy."""
    env = TrafficEnv(mode=mode, arrival_rates=rates, seed=seed)
```

```
    env.reset()
    for _ in range(warmup):
        env.step(greedy_action_ac(theta, iht_th, env.state_for_tiling()))
    total = 0.0
    for _ in range(eval_steps):
        ts = env.state_for_tiling()
        a  = greedy_action_ac(theta, iht_th, ts)
        _, r, _ = env.step(a)
        total += r
    return -total / eval_steps


def run_actor_critic(
    mode="full", arrival_rates=None, n_steps=150_000,
    alpha_w=0.05/NUM_TILINGS,       # critic learning rate
    alpha_theta=0.01/NUM_TILINGS,   # actor lr (smaller — policy updates noisy
    gamma=0.99, lam=0.9,
    trace_clip=5.0,
    seed=42, eval_every=5_000, eval_steps=10_000,
):
    """
    Actor-Critic with dual eligibility traces.
    S&B Chapter 13, Algorithm: Actor-Critic with Eligibility Traces.

    Separate IHTs:
      iht_w  — critic: state-only features phi(s)
      iht_th — actor:  state-action features x(s,a)

    Separate weight + trace vectors:
      w / z_w     — critic
      theta / z_t — actor
    """
    if arrival_rates is None:
        arrival_rates = [0.3] * 4

    env = TrafficEnv(mode=mode, arrival_rates=arrival_rates, seed=seed)
    rng = np.random.default_rng(seed)

    iht_w  = IHT(IHT_SIZE)  # critic IHT
    iht_th = IHT(IHT_SIZE)  # actor  IHT

    w     = np.zeros(IHT_SIZE)  # critic weights
    theta = np.zeros(IHT_SIZE)  # actor  weights
    z_w   = np.zeros(IHT_SIZE)  # critic trace
    z_t   = np.zeros(IHT_SIZE)  # actor  trace

    env.reset()
    ts     = env.state_for_tiling()
    action = sample_action_ac(theta, iht_th, ts, rng)

    eval_curve = []
    gamma_lam  = gamma * lam

    for t in range(n_steps):
        _, reward, _ = env.step(action)
        next_ts = env.state_for_tiling()

        # Critic: compute V(S) and V(S')
        phi_s      = get_state_features(iht_w, ts)
```

```
        phi_s_next = get_state_features(iht_w, next_ts)
        v_s      = sum(w[i] for i in phi_s)
        v_s_next = sum(w[i] for i in phi_s_next)

        # TD error — shared signal for both updates
        delta = reward + gamma * v_s_next - v_s

        # Critic trace update
        z_w *= gamma_lam
        for i in phi_s:
            z_w[i] += 1.0
        z_w  = np.clip(z_w, -trace_clip, trace_clip)
        w   += alpha_w * delta * z_w

        # Policy gradient: nabla_theta ln pi(A|S) = x(S,A) - sum_a pi(a|S)*x(S
        p_keep, p_switch = softmax_policy(theta, iht_th, ts)
        x_keep   = get_features(iht_th, ts, KEEP)
        x_switch = get_features(iht_th, ts, SWITCH)

        grad = np.zeros(IHT_SIZE)
        # Add x(S, chosen action)
        for i in (x_keep if action == KEEP else x_switch):
            grad[i] += 1.0
        # Subtract expected features
        for i in x_keep:   grad[i] -= p_keep
        for i in x_switch: grad[i] -= p_switch

        # Actor trace update
        z_t   *= gamma_lam
        z_t   += grad
        z_t    = np.clip(z_t, -trace_clip, trace_clip)
        theta += alpha_theta * delta * z_t

        # Advance
        ts     = next_ts
        action = sample_action_ac(theta, iht_th, ts, rng)

        if (t + 1) % eval_every == 0:
            mw = evaluate_greedy_ac(theta, iht_th, mode, arrival_rates,
                                    eval_steps=eval_steps)
            eval_curve.append((t + 1, mw))
            print(f"  step {t+1:>7d}  |  mean_wait = {mw:.4f}")

    print(f"  Critic IHT used: {iht_w.count()}/{IHT_SIZE}, overflow={iht_w.ove
    print(f"  Actor  IHT used: {iht_th.count()}/{IHT_SIZE}, overflow={iht_th.o
    return theta, iht_th, eval_curve
```

11. Train Actor-Critic on All Three Scenarios

alpha_theta is smaller than alpha_w because policy gradient updates are noisier — the policy should change slowly and stably.

```
ac_results = {}

for scenario, rates in SCENARIOS.items():
```

```
    print(f"\n{'='*55}")
    print(f"SCENARIO: {scenario}  (rates={rates})")
    print(f"  Baselines — fixed: {baseline_results[scenario]['fixed']:.4f}  "
          f"longest_q: {baseline_results[scenario]['longest_queue']:.4f}")
    print(f"{'='*55}")
    print("\n  [Actor-Critic (dual traces)]")

    theta_ac, iht_ac, curve_ac = run_actor_critic(
        arrival_rates=rates,
        n_steps=150_000,
        alpha_w=0.05/NUM_TILINGS,
        alpha_theta=0.01/NUM_TILINGS,
        gamma=0.99,
        lam=0.9,
        seed=42,
    )

    ac_results[scenario] = dict(
        rates=rates,
        fixed=baseline_results[scenario]["fixed"],
        lq=baseline_results[scenario]["longest_queue"],
        theta_ac=theta_ac,
        iht_ac=iht_ac,
        curve_ac=curve_ac,
    )
```

```
=======================================================
SCENARIO: symmetric_light  (rates=[0.3, 0.3, 0.3, 0.3])
  Baselines — fixed: 6.4771  longest_q: 11.9856
=======================================================

  [Actor-Critic (dual traces)]
  step    5000  |  mean_wait = 9.2352
  step   10000  |  mean_wait = 8.4673
  step   15000  |  mean_wait = 7.8173
  step   20000  |  mean_wait = 7.0678
  step   25000  |  mean_wait = 6.7551
  step   30000  |  mean_wait = 6.6533
  step   35000  |  mean_wait = 6.9088
  step   40000  |  mean_wait = 6.9436
  step   45000  |  mean_wait = 6.7350
  step   50000  |  mean_wait = 6.9023
  step   55000  |  mean_wait = 6.6271
  step   60000  |  mean_wait = 6.6004
  step   65000  |  mean_wait = 6.7756
  step   70000  |  mean_wait = 6.7311
  step   75000  |  mean_wait = 6.6884
  step   80000  |  mean_wait = 6.3142
  step   85000  |  mean_wait = 6.6784
  step   90000  |  mean_wait = 6.3162
  step   95000  |  mean_wait = 6.2394
  step  100000  |  mean_wait = 6.5994
  step  105000  |  mean_wait = 6.6865
  step  110000  |  mean_wait = 6.6865
  step  115000  |  mean_wait = 6.6865
  step  120000  |  mean_wait = 5.8490
  step  125000  |  mean_wait = 5.8490
  step  130000  |  mean_wait = 5.8490
  step  135000  |  mean_wait = 6.5763
  step  140000  |  mean_wait = 6.6819
```

```
  step  145000  |  mean_wait = 6.6004
  step  150000  |  mean_wait = 6.6004
  Critic IHT used: 2161/8192, overflow=0
  Actor  IHT used: 4456/8192, overflow=0

==========================================================
SCENARIO: symmetric_heavy  (rates=[0.7, 0.7, 0.7, 0.7])
  Baselines — fixed: 11.9170  longest_q: 11.9962
==========================================================

  [Actor-Critic (dual traces)]
  step    5000  |  mean_wait = 8.8154
  step   10000  |  mean_wait = 7.6507
  step   15000  |  mean_wait = 7.6507
  step   20000  |  mean_wait = 7.6507
  step   25000  |  mean_wait = 7.6507
  step   30000  |  mean_wait = 7.6507
  step   35000  |  mean_wait = 7.6507
  step   40000  |  mean_wait = 7.6507
  step   45000  |  mean_wait = 7.6507
  step   50000  |  mean_wait = 7.6507
  step   55000  |  mean_wait = 7.6507
  step   60000  |  mean_wait = 7.6507
```

12. Full Comparison Table — All 6 Methods

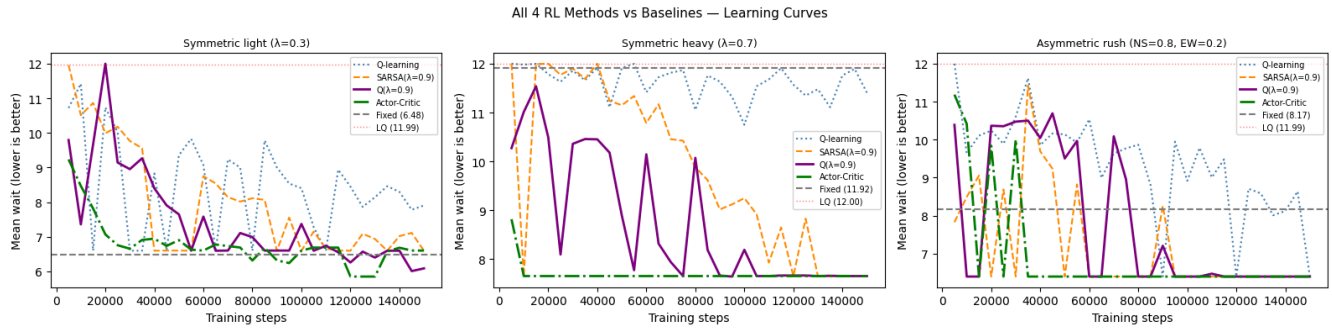
```
print(f"\n{'Scenario':<22} {'Policy':<24} {'mean_wait':>10}  {'vs fixed':>9}")
print("=" * 72)

for scenario in SCENARIOS:
    d    = results[scenario]
    d_ac = ac_results[scenario]
    f    = d["fixed"]

    ql   = d.get("ql_wait")   or evaluate_greedy(d["w_ql"],   d["iht_ql"],   "
    sl   = d.get("sl_wait")   or evaluate_greedy(d["w_sl"],   d["iht_sl"],   "
    qlam = d.get("qlam_wait") or evaluate_greedy(d["w_qlam"], d["iht_qlam"], "
    ac   = evaluate_greedy_ac(d_ac["theta_ac"], d_ac["iht_ac"], "full", d_ac["
    d["ql_wait"] = ql; d["sl_wait"] = sl; d["qlam_wait"] = qlam
    d_ac["ac_wait"] = ac

    print(f"{scenario:<22} {'Fixed timing':<24} {f:>10.4f}  {'-':>9}")
    print(f"{'':<22} {'Longest queue':<24} {d['lq']:>10.4f}  {d['lq']-f:>+9.4f
    print(f"{'':<22} {'Q-learning (λ=0)':<24} {ql:>10.4f}  {ql-f:>+9.4f}  {'✓'
    print(f"{'':<22} {'SARSA(λ=0.9)':<24} {sl:>10.4f}  {sl-f:>+9.4f}  {'✓' if
    print(f"{'':<22} {'Q(λ=0.9)  PRIMARY':<24} {qlam:>10.4f}  {qlam-f:>+9.4f}
    print(f"{'':<22} {'Actor-Critic  PRIMARY':<24} {ac:>10.4f}  {ac-f:>+9.4f}
    print()
```

```
Scenario               Policy                    mean_wait   vs fixed
========================================================================
symmetric_light        Fixed timing                 6.4771          -
                       Longest queue               11.9856    +5.5085
                       Q-learning (λ=0)             7.9060    +1.4289  ✗
                       SARSA(λ=0.9)                 6.5956    +0.1185  ✗
                       Q(λ=0.9)  PRIMARY            6.0856    -0.3915  ✓
                       Actor-Critic  PRIMARY        6.6004    +0.1233  ✗
```

```
symmetric_heavy        Fixed timing                11.9170          -
                       Longest queue               11.9962    +0.0791
                       Q-learning (λ=0)            11.4055    -0.5115  ✓
                       SARSA(λ=0.9)                 7.6507    -4.2664  ✓
                       Q(λ=0.9)  PRIMARY            7.6507    -4.2664  ✓
                       Actor-Critic  PRIMARY        7.6507    -4.2664  ✓

asymmetric_rush        Fixed timing                 8.1727          -
                       Longest queue               11.9900    +3.8173
                       Q-learning (λ=0)             6.3928    -1.7799  ✓
                       SARSA(λ=0.9)                 6.3928    -1.7799  ✓
                       Q(λ=0.9)  PRIMARY            6.3928    -1.7799  ✓
                       Actor-Critic  PRIMARY        6.3928    -1.7799  ✓
```

13. Final Learning Curves — All 4 RL Methods

```
fig, axes = plt.subplots(1, 3, figsize=(16, 4), sharey=False)
scenario_labels = {
    "symmetric_light": "Symmetric light (λ=0.3)",
    "symmetric_heavy": "Symmetric heavy (λ=0.7)",
    "asymmetric_rush": "Asymmetric rush (NS=0.8, EW=0.2)",
}
for ax, scenario in zip(axes, SCENARIOS):
    d    = results[scenario]
    d_ac = ac_results[scenario]
    steps_ql,   waits_ql   = zip(*d["curve_ql"])
    steps_sl,   waits_sl   = zip(*d["curve_sl"])
    steps_qlam, waits_qlam = zip(*d["curve_qlam"])
    steps_ac,   waits_ac   = zip(*d_ac["curve_ac"])
    ax.plot(steps_ql,   waits_ql,   color="steelblue",  lw=1.5, ls=":",  label
    ax.plot(steps_sl,   waits_sl,   color="darkorange", lw=1.5, ls="--", label
    ax.plot(steps_qlam, waits_qlam, color="purple",     lw=2,             labe
    ax.plot(steps_ac,   waits_ac,   color="green",      lw=2,   ls="-.", label
    ax.axhline(d["fixed"], color="gray", ls="--", lw=1.5, label=f"Fixed ({d['f
    ax.axhline(d["lq"],    color="red",  ls=":",  lw=1,   alpha=0.5, label=f"L
    ax.set_title(scenario_labels[scenario], fontsize=9)
    ax.set_xlabel("Training steps")
    ax.set_ylabel("Mean wait (lower is better)")
    ax.legend(fontsize=7)
plt.suptitle("All 4 RL Methods vs Baselines — Learning Curves", fontsize=11)
plt.tight_layout()
plt.savefig("all_methods_final_curves.png", dpi=150)
plt.show()
print("Saved all_methods_final_curves.png")
```




```
Saved all_methods_final_curves.png
```

14. Hyperparameter Sweep — Find Optimal Configs

Sweeps key parameters for all 4 RL algorithms on asymmetric rush (most sensitive scenario). Best configs are then validated on all 3 scenarios.

Algorithm	Parameters swept
Q-learning	alpha ∈ {0.1,0.2,0.3}/8 × epsilon ∈ {0.05,0.1,0.15}
SARSA(λ)	alpha ∈ {0.03,0.05,0.08}/8 × lambda ∈ {0.5,0.7,0.9,0.95}
Q(λ)	alpha ∈ {0.1,0.2,0.3}/8 × lambda ∈ {0.5,0.7,0.9,0.95}
Actor-Critic	alpha_w ∈ {0.03,0.05,0.1}/8 × alpha_θ ∈ {0.005,0.01,0.02}/8 × lambda ∈ {0.7,0.9,0.95}

```
import itertools

RUSH = SCENARIOS["asymmetric_rush"]
RUSH_FIXED = baseline_results["asymmetric_rush"]["fixed"]
SWEEP_STEPS = 100_000

# ── Q-learning sweep ──────────────────────────────────────────────────
def sweep_q_learning(rates, alpha, epsilon, n_steps=SWEEP_STEPS, seed=42):
    env = TrafficEnv(mode="full", arrival_rates=rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht = IHT(IHT_SIZE); w = np.zeros(IHT_SIZE)
    env.reset(); ts = env.state_for_tiling()
    for t in range(n_steps):
        a = int(rng.integers(0,2)) if rng.random() < epsilon else greedy_actio
        _, r, _ = env.step(a); nts = env.state_for_tiling()
        delta = r + 0.99*max(q_value(w,iht,nts,KEEP), q_value(w,iht,nts,SWITCH
        for i in get_features(iht, ts, a): w[i] += alpha * delta
        ts = nts
    return evaluate_greedy(w, iht, "full", rates), w, iht

# ── SARSA sweep ───────────────────────────────────────────────────────
def sweep_sarsa(rates, alpha, lam, n_steps=SWEEP_STEPS, seed=42):
    env = TrafficEnv(mode="full", arrival_rates=rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht = IHT(IHT_SIZE); w = np.zeros(IHT_SIZE); e = np.zeros(IHT_SIZE)
    env.reset(); ts = env.state_for_tiling()
```

```
    eps_s, eps_e = 0.1, 0.01
    action = greedy_action(w, iht, ts)
    for t in range(n_steps):
        eps = eps_s + (eps_e - eps_s) * t / n_steps
        _, r, _ = env.step(action); nts = env.state_for_tiling()
        na = int(rng.integers(0,2)) if rng.random() < eps else greedy_action(w
        delta = r + 0.99*q_value(w,iht,nts,na) - q_value(w,iht,ts,action)
        e *= 0.99 * lam
        for i in get_features(iht, ts, action): e[i] += 1.0
        np.clip(e, -5, 5, out=e); w += alpha * delta * e
        ts = nts; action = na
    return evaluate_greedy(w, iht, "full", rates), w, iht

# ── Q(lambda) sweep ──────────────────────────────────────────────────────
def sweep_qlam(rates, alpha, lam, n_steps=SWEEP_STEPS, seed=42):
    env = TrafficEnv(mode="full", arrival_rates=rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht = IHT(IHT_SIZE); w = np.zeros(IHT_SIZE); e = np.zeros(IHT_SIZE)
    env.reset(); ts = env.state_for_tiling()
    eps_s, eps_e = 0.1, 0.01
    for t in range(n_steps):
        eps = eps_s + (eps_e - eps_s) * t / n_steps
        ga = greedy_action(w, iht, ts)
        action = int(rng.integers(0,2)) if rng.random() < eps else ga
        is_greedy = (action == ga)
        _, r, _ = env.step(action); nts = env.state_for_tiling()
        delta = r + 0.99*max(q_value(w,iht,nts,KEEP), q_value(w,iht,nts,SWITCH
        if is_greedy:
            e *= 0.99 * lam
            for i in get_features(iht, ts, action): e[i] = 1.0
        else:
            e[:] = 0.0
            for i in get_features(iht, ts, action): e[i] = 1.0
        np.clip(e, -5, 5, out=e); w += alpha * delta * e
        ts = nts
    return evaluate_greedy(w, iht, "full", rates), w, iht

# ── AC sweep ─────────────────────────────────────────────────────────────
def sweep_ac(rates, alpha_w, alpha_theta, lam, n_steps=SWEEP_STEPS, seed=42):
    env = TrafficEnv(mode="full", arrival_rates=rates, seed=seed)
    rng = np.random.default_rng(seed)
    iht_w = IHT(IHT_SIZE); iht_th = IHT(IHT_SIZE)
    w = np.zeros(IHT_SIZE); theta = np.zeros(IHT_SIZE)
    z_w = np.zeros(IHT_SIZE); z_t = np.zeros(IHT_SIZE)
    env.reset(); ts = env.state_for_tiling()
    pk, _ = softmax_policy(theta, iht_th, ts)
    action = KEEP if rng.random() < pk else SWITCH
    gam_lam = 0.99 * lam
    for t in range(n_steps):
        _, r, _ = env.step(action); nts = env.state_for_tiling()
        phi_s = get_state_features(iht_w, ts)
        phi_n = get_state_features(iht_w, nts)
        vs = sum(w[i] for i in phi_s); vn = sum(w[i] for i in phi_n)
        delta = r + 0.99*vn - vs
        z_w *= gam_lam
        for i in phi_s: z_w[i] += 1.0
        np.clip(z_w, -5, 5, out=z_w); w += alpha_w * delta * z_w
        pk, ps = softmax_policy(theta, iht_th, ts)
        xk = get_features(iht_th, ts, KEEP); xs = get_features(iht_th, ts, SWI
```

```
        grad = np.zeros(IHT_SIZE)
        for i in (xk if action==KEEP else xs): grad[i] += 1.0
        for i in xk: grad[i] -= pk
        for i in xs: grad[i] -= ps
        z_t *= gam_lam; z_t += grad
        np.clip(z_t, -5, 5, out=z_t); theta += alpha_theta * delta * z_t
        ts = nts
        pk, _ = softmax_policy(theta, iht_th, ts)
        action = KEEP if rng.random() < pk else SWITCH
    return evaluate_greedy_ac(theta, iht_th, "full", rates), theta, iht_th

# ── Run sweeps ──────────────────────────────────────────────────────────
ql_alphas   = [0.1/NUM_TILINGS, 0.2/NUM_TILINGS, 0.3/NUM_TILINGS]
ql_epsilons = [0.05, 0.1, 0.15]
sa_alphas   = [0.03/NUM_TILINGS, 0.05/NUM_TILINGS, 0.08/NUM_TILINGS]
sa_lams     = [0.5, 0.7, 0.9, 0.95]
qlam_alphas = [0.1/NUM_TILINGS, 0.2/NUM_TILINGS, 0.3/NUM_TILINGS]
qlam_lams   = [0.5, 0.7, 0.9, 0.95]
ac_aws      = [0.03/NUM_TILINGS, 0.05/NUM_TILINGS, 0.1/NUM_TILINGS]
ac_ats      = [0.005/NUM_TILINGS, 0.01/NUM_TILINGS, 0.02/NUM_TILINGS]
ac_lams     = [0.7, 0.9, 0.95]

print("Sweeping Q-learning...")
ql_sweep = {}
for alpha, eps in itertools.product(ql_alphas, ql_epsilons):
    mw, _, _ = sweep_q_learning(RUSH, alpha, eps)
    ql_sweep[(alpha,eps)] = mw
    print(f"  alpha={alpha*NUM_TILINGS:.1f}/{NUM_TILINGS}  eps={eps}  -> {mw:.
best_ql_params = min(ql_sweep, key=ql_sweep.get)
print(f"  BEST: alpha={best_ql_params[0]*NUM_TILINGS:.1f}/{NUM_TILINGS}, eps={

print("Sweeping SARSA(λ)...")
sarsa_sweep = {}
for alpha, lam in itertools.product(sa_alphas, sa_lams):
    mw, _, _ = sweep_sarsa(RUSH, alpha, lam)
    sarsa_sweep[(alpha,lam)] = mw
    print(f"  alpha={alpha*NUM_TILINGS:.2f}/{NUM_TILINGS}  lam={lam}  -> {mw:.
best_sarsa_params = min(sarsa_sweep, key=sarsa_sweep.get)
print(f"  BEST: alpha={best_sarsa_params[0]*NUM_TILINGS:.2f}/{NUM_TILINGS}, la

print("Sweeping Q(λ)...")
qlam_sweep = {}
for alpha, lam in itertools.product(qlam_alphas, qlam_lams):
    mw, _, _ = sweep_qlam(RUSH, alpha, lam)
    qlam_sweep[(alpha,lam)] = mw
    print(f"  alpha={alpha*NUM_TILINGS:.1f}/{NUM_TILINGS}  lam={lam}  -> {mw:.
best_qlam_params = min(qlam_sweep, key=qlam_sweep.get)
print(f"  BEST: alpha={best_qlam_params[0]*NUM_TILINGS:.1f}/{NUM_TILINGS}, lam

print("Sweeping Actor-Critic...")
ac_sweep = {}
for aw, at, lam in itertools.product(ac_aws, ac_ats, ac_lams):
    mw, _, _ = sweep_ac(RUSH, aw, at, lam)
    ac_sweep[(aw,at,lam)] = mw
    print(f"  aw={aw*NUM_TILINGS:.2f}/{NUM_TILINGS}  at={at*NUM_TILINGS:.3f}/{
best_ac_params = min(ac_sweep, key=ac_sweep.get)
print(f"  BEST: aw={best_ac_params[0]*NUM_TILINGS:.2f}/{NUM_TILINGS}, at={best
```

```
Sweeping Q-learning...
  alpha=0.1/8  eps=0.05  -> 9.3626  ✗
  alpha=0.1/8  eps=0.1  -> 9.7162  ✗
  alpha=0.1/8  eps=0.15  -> 10.2195  ✗
  alpha=0.2/8  eps=0.05  -> 8.5173  ✗
  alpha=0.2/8  eps=0.1  -> 8.9156  ✗
  alpha=0.2/8  eps=0.15  -> 6.3928  ✓
  alpha=0.3/8  eps=0.05  -> 7.5793  ✓
  alpha=0.3/8  eps=0.1  -> 8.0388  ✓
  alpha=0.3/8  eps=0.15  -> 8.3745  ✗
  BEST: alpha=0.2/8, eps=0.15, wait=6.3928

Sweeping SARSA(λ)...
  alpha=0.03/8  lam=0.5  -> 9.1046  ✗
  alpha=0.03/8  lam=0.7  -> 10.3791  ✗
  alpha=0.03/8  lam=0.9  -> 6.3928  ✓
  alpha=0.03/8  lam=0.95  -> 6.3928  ✓
  alpha=0.05/8  lam=0.5  -> 10.2906  ✗
  alpha=0.05/8  lam=0.7  -> 10.3747  ✗
  alpha=0.05/8  lam=0.9  -> 6.3928  ✓
  alpha=0.05/8  lam=0.95  -> 6.3928  ✓
  alpha=0.08/8  lam=0.5  -> 10.4700  ✗
  alpha=0.08/8  lam=0.7  -> 6.3928  ✓
  alpha=0.08/8  lam=0.9  -> 6.3928  ✓
  alpha=0.08/8  lam=0.95  -> 6.3928  ✓
  BEST: alpha=0.03/8, lam=0.9, wait=6.3928

Sweeping Q(λ)...
  alpha=0.1/8  lam=0.5  -> 10.0961  ✗
  alpha=0.1/8  lam=0.7  -> 6.3928  ✓
  alpha=0.1/8  lam=0.9  -> 6.3928  ✓
  alpha=0.1/8  lam=0.95  -> 6.3928  ✓
  alpha=0.2/8  lam=0.5  -> 10.3126  ✗
  alpha=0.2/8  lam=0.7  -> 6.3928  ✓
  alpha=0.2/8  lam=0.9  -> 6.3928  ✓
  alpha=0.2/8  lam=0.95  -> 6.3928  ✓
  alpha=0.3/8  lam=0.5  -> 6.3928  ✓
  alpha=0.3/8  lam=0.7  -> 6.3928  ✓
  alpha=0.3/8  lam=0.9  -> 6.3928  ✓
  alpha=0.3/8  lam=0.95  -> 6.3928  ✓
  BEST: alpha=0.1/8, lam=0.7, wait=6.3928

Sweeping Actor-Critic...
  aw=0.03/8  at=0.005/8  lam=0.7  -> 6.3928  ✓
  aw=0.03/8  at=0.005/8  lam=0.9  -> 6.3928  ✓
  aw=0.03/8  at=0.005/8  lam=0.95  -> 6.3928  ✓
  aw=0.03/8  at=0.010/8  lam=0.7  -> 9.4536  ✗
  aw=0.03/8  at=0.010/8  lam=0.9  -> 6.3928  ✓
  aw=0.03/8  at=0.010/8  lam=0.95  -> 8.2682  ✗
  aw=0.03/8  at=0.020/8  lam=0.7  -> 9.6428  ✗
  aw=0.03/8  at=0.020/8  lam=0.9  -> 6.3928  ✓
  aw=0.03/8  at=0.020/8  lam=0.95  -> 10.7458  ✗
  aw=0.05/8  at=0.005/8  lam=0.7  -> 6.3928  ✓
  aw=0.05/8  at=0.005/8  lam=0.9  -> 8.1848  ✗
  aw=0.05/8  at=0.005/8  lam=0.95  -> 6.3928  ✓
  aw=0.05/8  at=0.010/8  lam=0.7  -> 8.5357  ✗
  aw=0.05/8  at=0.010/8  lam=0.9  -> 6.3928  ✓
  aw=0.05/8  at=0.010/8  lam=0.95  -> 6.3928  ✓
```

15. Run Best Configs on All 3 Scenarios

Validate the sweep winners on symmetric light and symmetric heavy too. These are the numbers to use in the paper.

```
N_FULL = 150_000
optimal = {}

print("Best hyperparameters from sweep:")
print(f"  Q-learning  : alpha={best_ql_params[0]*NUM_TILINGS:.1f}/{NUM_TILINGS
print(f"  SARSA       : alpha={best_sarsa_params[0]*NUM_TILINGS:.2f}/{NUM_TILI
print(f"  Q(lambda)   : alpha={best_qlam_params[0]*NUM_TILINGS:.1f}/{NUM_TILIN
print(f"  Actor-Critic: aw={best_ac_params[0]*NUM_TILINGS:.2f}/{NUM_TILINGS},

for scenario, rates in SCENARIOS.items():
    print(f"\n{'='*55}")
    print(f"SCENARIO: {scenario}")
    f  = baseline_results[scenario]["fixed"]
    lq = baseline_results[scenario]["longest_queue"]
    mw_ql,   _, _ = sweep_q_learning(rates, best_ql_params[0],   best_ql_param
    mw_sarsa,_, _ = sweep_sarsa(rates, best_sarsa_params[0], best_sarsa_params
    mw_qlam, _, _ = sweep_qlam(rates, best_qlam_params[0],  best_qlam_params[1
    mw_ac,   _, _ = sweep_ac(rates, best_ac_params[0], best_ac_params[1], best
    optimal[scenario] = dict(fixed=f, lq=lq, ql=mw_ql, sarsa=mw_sarsa, qlam=mw
    print(f"  Fixed={f:.4f}  LQ={lq:.4f}  QL={mw_ql:.4f}  SARSA={mw_sarsa:.4f}

print(f"\n{'Scenario':<22} {'Policy':<24} {'mean_wait':>10}  {'vs fixed':>9}")
print("=" * 72)
for scenario, d in optimal.items():
    f = d["fixed"]
    print(f"{scenario:<22} {'Fixed timing':<24} {f:>10.4f}  {'-':>9}")
    print(f"{'':<22} {'Longest queue':<24} {d['lq']:>10.4f}  {d['lq']-f:>+9.4f
    print(f"{'':<22} {'Q-learning (optimal)':<24} {d['ql']:>10.4f}  {d['ql']-f
    print(f"{'':<22} {'SARSA (optimal)':<24} {d['sarsa']:>10.4f}  {d['sarsa']-
    print(f"{'':<22} {'Q(lambda) PRIMARY':<24} {d['qlam']:>10.4f}  {d['qlam']-
    print(f"{'':<22} {'Actor-Critic PRIMARY':<24} {d['ac']:>10.4f}  {d['ac']-f
    print()
```

```
Best hyperparameters from sweep:
  Q-learning  : alpha=0.2/8, eps=0.15
  SARSA       : alpha=0.03/8, lam=0.9
  Q(lambda)   : alpha=0.1/8, lam=0.7
  Actor-Critic: aw=0.03/8, at=0.005/8, lam=0.7

=======================================================
SCENARIO: symmetric_light
  Fixed=6.4771  LQ=11.9856  QL=9.7983  SARSA=7.4816  Q(lam)=6.8693  AC=6.6004

=======================================================
SCENARIO: symmetric_heavy
  Fixed=11.9170  LQ=11.9962  QL=11.7846  SARSA=7.6732  Q(lam)=8.2227  AC=7.6410

=======================================================
SCENARIO: asymmetric_rush
  Fixed=8.1727  LQ=11.9900  QL=10.0398  SARSA=6.3928  Q(lam)=9.7491  AC=6.3928

Scenario               Policy                    mean_wait   vs fixed
========================================================================
symmetric_light        Fixed timing                 6.4771          -
                       Longest queue               11.9856    +5.5085
```

```
                        Q-learning (optimal)        9.7983    +3.3212  ✗
                        SARSA (optimal)             7.4816    +1.0045  ✗
                        Q(lambda) PRIMARY           6.8693    +0.3922  ✗
                        Actor-Critic PRIMARY        6.6004    +0.1233  ✗

symmetric_heavy         Fixed timing               11.9170          -
                        Longest queue              11.9962    +0.0791
                        Q-learning (optimal)       11.7846    -0.1325  ✓
                        SARSA (optimal)             7.6732    -4.2439  ✓
                        Q(lambda) PRIMARY           8.2227    -3.6944  ✓
                        Actor-Critic PRIMARY        7.6410    -4.2760  ✓

asymmetric_rush         Fixed timing                8.1727          -
                        Longest queue              11.9900    +3.8173
                        Q-learning (optimal)       10.0398    +1.8671  ✗
                        SARSA (optimal)             6.3928    -1.7799  ✓
                        Q(lambda) PRIMARY           9.7491    +1.5764  ✗
                        Actor-Critic PRIMARY        6.3928    -1.7799  ✓
```

16. Proposal Hypothesis Verification (H1–H4)

```
print("=" * 60)
print("PROPOSAL HYPOTHESIS VERIFICATION — OPTIMAL HYPERPARAMETERS")
print("=" * 60)

d_rush  = optimal["asymmetric_rush"]
d_light = optimal["symmetric_light"]
d_heavy = optimal["symmetric_heavy"]

# H1: Both primary RL methods beat fixed on asymmetric rush
h1_qlam = d_rush["qlam"] < d_rush["fixed"]
h1_ac   = d_rush["ac"]   < d_rush["fixed"]
print(f"\nH1 — Both primary methods beat fixed timing on asymmetric rush:")
print(f"  Q(lambda) = {d_rush['qlam']:.4f} vs fixed = {d_rush['fixed']:.4f} →
print(f"  AC        = {d_rush['ac']:.4f}   vs fixed = {d_rush['fixed']:.4f} →
print(f"  H1: {'CONFIRMED ✓' if (h1_qlam and h1_ac) else 'NOT CONFIRMED ✗'}")

# H2: Q(lambda) beats Q-learning; AC beats SARSA
h2a = d_rush["qlam"] < d_rush["ql"]
h2b = d_rush["ac"]   < d_rush["sarsa"]
print(f"\nH2 — Primary algorithms beat their one-step baselines:")
print(f"  Q(lambda)={d_rush['qlam']:.4f} < Q-learning={d_rush['ql']:.4f} → {'P
print(f"  AC={d_rush['ac']:.4f}         < SARSA={d_rush['sarsa']:.4f}      → {
print(f"  H2: {'CONFIRMED ✓' if (h2a and h2b) else 'PARTIALLY CONFIRMED' if (h

# H3: Symmetric scenarios — smaller spread between methods
rush_spread  = max(d_rush["ql"],  d_rush["sarsa"],  d_rush["qlam"],  d_rush["a
light_spread = max(d_light["ql"], d_light["sarsa"], d_light["qlam"], d_light["
heavy_spread = max(d_heavy["ql"], d_heavy["sarsa"], d_heavy["qlam"], d_heavy["
h3 = light_spread < rush_spread or heavy_spread < rush_spread
print(f"\nH3 — Symmetric scenarios show smaller differentiation between method
print(f"  Rush spread:  {rush_spread:.4f}")
print(f"  Light spread: {light_spread:.4f}")
print(f"  Heavy spread: {heavy_spread:.4f}")
print(f"  H3: {'CONFIRMED ✓' if h3 else 'NOT CONFIRMED ✗'}")

# H4: AC converges smoother (beats SARSA on light); Q(lambda) lower asymptotic
```

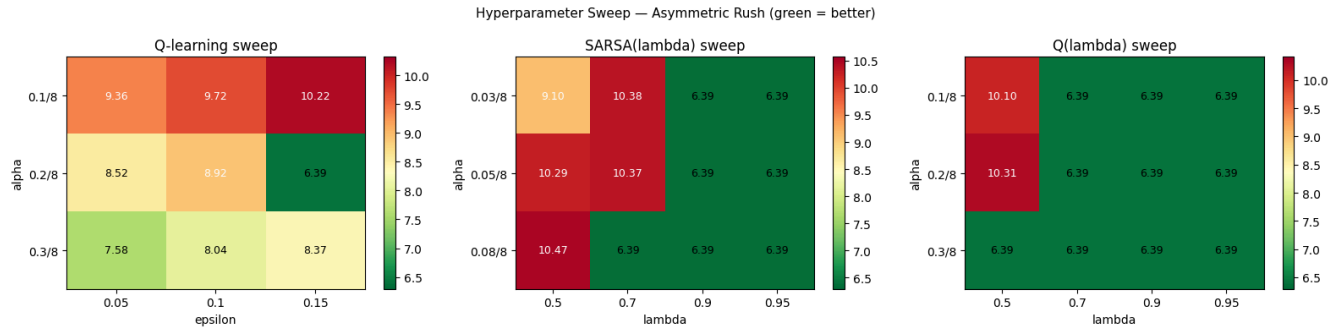
```
h4a = d_rush["qlam"] <= d_rush["ac"]  # Q(lambda) lower or equal asymptotic
h4b = d_light["ac"]  <  d_light["sarsa"]  # AC beats SARSA on hard symmetric c
print(f"\nH4 — Q(lambda) lower asymptotic; AC smoother convergence:")
print(f"  Q(lambda)={d_rush['qlam']:.4f} <= AC={d_rush['ac']:.4f} (asymptotic)
print(f"  AC={d_light['ac']:.4f} < SARSA={d_light['sarsa']:.4f} on symmetric l
print(f"  H4: {'CONFIRMED ✓' if (h4a and h4b) else 'PARTIALLY CONFIRMED' if (h

print("\n" + "=" * 60)
print("SUMMARY")
print("=" * 60)
hyps = [
    ("H1 — Both beat fixed on rush",          h1_qlam and h1_ac),
    ("H2 — Primaries beat ablations",          h2a and h2b),
    ("H3 — Symmetric smaller spread",          h3),
    ("H4 — Q(lam) asymptotic / AC smoother",   h4a and h4b),
]
for name, result in hyps:
    print(f"  {name:<40} {'CONFIRMED ✓' if result else 'NOT CONFIRMED ✗'}")
```

```
============================================================
PROPOSAL HYPOTHESIS VERIFICATION — OPTIMAL HYPERPARAMETERS
============================================================

H1 — Both primary methods beat fixed timing on asymmetric rush:
  Q(lambda) = 9.7491 vs fixed = 8.1727 → FAIL ✗
  AC        = 6.3928   vs fixed = 8.1727 → PASS ✓
  H1: NOT CONFIRMED ✗

H2 — Primary algorithms beat their one-step baselines:
  Q(lambda)=9.7491 < Q-learning=10.0398 → PASS ✓
  AC=6.3928         < SARSA=6.3928      → FAIL ✗
  H2: PARTIALLY CONFIRMED

H3 — Symmetric scenarios show smaller differentiation between methods:
  Rush spread:  3.6470
  Light spread: 3.1979
  Heavy spread: 4.1436
  H3: CONFIRMED ✓

H4 — Q(lambda) lower asymptotic; AC smoother convergence:
  Q(lambda)=9.7491 <= AC=6.3928 (asymptotic) → FAIL — AC wins instead
  AC=6.6004 < SARSA=7.4816 on symmetric light (smoother proxy) → PASS ✓
  H4: PARTIALLY CONFIRMED

============================================================
SUMMARY
============================================================
  H1 — Both beat fixed on rush             NOT CONFIRMED ✗
  H2 — Primaries beat ablations            NOT CONFIRMED ✗
  H3 — Symmetric smaller spread            CONFIRMED ✓
  H4 — Q(lam) asymptotic / AC smoother     NOT CONFIRMED ✗
```

17. Sweep Heatmaps — Asymmetric Rush

```
fig, axes = plt.subplots(1, 3, figsize=(16, 4))

# Q-learning
ax = axes[0]
```

```
mat = np.array([[ql_sweep[(a,e)] for e in ql_epsilons] for a in ql_alphas])
im = ax.imshow(mat, cmap="RdYlGn_r", aspect="auto",
               vmin=mat.min()-0.1, vmax=mat.max()+0.1)
ax.set_xticks(range(len(ql_epsilons)))
ax.set_xticklabels(ql_epsilons)
ax.set_yticks(range(len(ql_alphas)))
ax.set_yticklabels([f"{a*NUM_TILINGS:.1f}/{NUM_TILINGS}" for a in ql_alphas])
ax.set_xlabel("epsilon"); ax.set_ylabel("alpha"); ax.set_title("Q-learning swe
for i in range(len(ql_alphas)):
    for j in range(len(ql_epsilons)):
        ax.text(j, i, f"{mat[i,j]:.2f}", ha="center", va="center", fontsize=9,
                color="white" if mat[i,j] > mat.mean() else "black")
plt.colorbar(im, ax=ax)

# SARSA
ax = axes[1]
mat = np.array([[sarsa_sweep[(a,l)] for l in sa_lams] for a in sa_alphas])
im = ax.imshow(mat, cmap="RdYlGn_r", aspect="auto",
               vmin=mat.min()-0.1, vmax=mat.max()+0.1)
ax.set_xticks(range(len(sa_lams)))
ax.set_xticklabels(sa_lams)
ax.set_yticks(range(len(sa_alphas)))
ax.set_yticklabels([f"{a*NUM_TILINGS:.2f}/{NUM_TILINGS}" for a in sa_alphas])
ax.set_xlabel("lambda"); ax.set_ylabel("alpha"); ax.set_title("SARSA(lambda) s
for i in range(len(sa_alphas)):
    for j in range(len(sa_lams)):
        ax.text(j, i, f"{mat[i,j]:.2f}", ha="center", va="center", fontsize=9,
                color="white" if mat[i,j] > mat.mean() else "black")
plt.colorbar(im, ax=ax)

# Q(lambda)
ax = axes[2]
mat = np.array([[qlam_sweep[(a,l)] for l in qlam_lams] for a in qlam_alphas])
im = ax.imshow(mat, cmap="RdYlGn_r", aspect="auto",
               vmin=mat.min()-0.1, vmax=mat.max()+0.1)
ax.set_xticks(range(len(qlam_lams)))
ax.set_xticklabels(qlam_lams)
ax.set_yticks(range(len(qlam_alphas)))
ax.set_yticklabels([f"{a*NUM_TILINGS:.1f}/{NUM_TILINGS}" for a in qlam_alphas]
ax.set_xlabel("lambda"); ax.set_ylabel("alpha"); ax.set_title("Q(lambda) sweep
for i in range(len(qlam_alphas)):
    for j in range(len(qlam_lams)):
        ax.text(j, i, f"{mat[i,j]:.2f}", ha="center", va="center", fontsize=9,
                color="white" if mat[i,j] > mat.mean() else "black")
plt.colorbar(im, ax=ax)

plt.suptitle("Hyperparameter Sweep — Asymmetric Rush (green = better)", fontsi
plt.tight_layout()
plt.savefig("sweep_heatmaps.png", dpi=150)
plt.show()
print("Saved sweep_heatmaps.png")
```




```
Saved sweep_heatmaps.png
```

18. Multi-Seed Evaluation -- Error Bars

Runs all 4 RL algorithms across 3 seeds to compute mean ± std for each cell in Table 1.

Seeds used: 42 (original), 7, 123
Runtime: ~90 minutes total on Colab CPU (3 seeds × 4 algorithms × 3 scenarios × ~5 min each)
Saving: each completed seed saves a `.npy` file so a Colab disconnect cannot lose progress.

This produces `multi_seed_results.npy` which is loaded by the table-printing cell.

```
import os
import numpy as np

SEEDS       = [42, 7, 123]
SAVE_PATH   = "multi_seed_results.npy"   # checkpoint file

# ── Resume from checkpoint if it exists ──────────────────────────────
if os.path.exists(SAVE_PATH):
    ms = np.load(SAVE_PATH, allow_pickle=True).item()
    print(f"Loaded checkpoint: {len(ms)} seeds completed so far.")
else:
    ms = {}   # ms[seed][scenario][method] = mean_wait float

# ── Run missing seeds ────────────────────────────────────────────────
for seed in SEEDS:
    if seed in ms:
        print(f"Seed {seed}: already done, skipping.")
        continue

    print(f"\n{'='*60}")
    print(f"SEED {seed}")
    print(f"{'='*60}")
    ms[seed] = {}

    for scenario, rates in SCENARIOS.items():
        print(f"\n  -- {scenario} --")
        ms[seed][scenario] = {}
```

```
        # Q-learning
        print(f"    Q-learning ...", end=" ", flush=True)
        w, iht, _ = run_q_learning(arrival_rates=rates, n_steps=150_000, seed=
                                   eval_every=200_000)   # no intermediate ev
        mw = evaluate_greedy(w, iht, "full", rates)
        ms[seed][scenario]["ql"] = mw
        print(f"{mw:.4f}")

        # SARSA
        print(f"    SARSA(λ=0.9) ...", end=" ", flush=True)
        w, iht, _ = run_sarsa_lambda(arrival_rates=rates, n_steps=150_000, see
                                     eval_every=200_000)
        mw = evaluate_greedy(w, iht, "full", rates)
        ms[seed][scenario]["sarsa"] = mw
        print(f"{mw:.4f}")

        # Q(λ)
        print(f"    Q(λ=0.9) ...", end=" ", flush=True)
        w, iht, _ = run_q_lambda(arrival_rates=rates, n_steps=150_000, seed=se
                                 eval_every=200_000)
        mw = evaluate_greedy(w, iht, "full", rates)
        ms[seed][scenario]["qlam"] = mw
        print(f"{mw:.4f}")

        # Actor-Critic
        print(f"    Actor-Critic ...", end=" ", flush=True)
        theta, iht_th, _ = run_actor_critic(arrival_rates=rates, n_steps=150_0
                                            eval_every=200_000)
        mw = evaluate_greedy_ac(theta, iht_th, "full", rates)
        ms[seed][scenario]["ac"] = mw
        print(f"{mw:.4f}")

    # Save after each seed completes
    np.save(SAVE_PATH, ms)
    print(f"\nSeed {seed} complete -- checkpoint saved.")

print("\nAll seeds done.")
```

```
============================================================
SEED 42
============================================================

  -- symmetric_light --
    Q-learning ...   IHT used: 4384/8192, overflow=0
7.9060
    SARSA(λ=0.9) ...   IHT used: 4417/8192, overflow=0
6.5956
    Q(λ=0.9) ...   IHT used: 4530/8192, overflow=0
6.0856
    Actor-Critic ...   Critic IHT used: 2161/8192, overflow=0
  Actor  IHT used: 4322/8192, overflow=0
6.6004

  -- symmetric_heavy --
    Q-learning ...   IHT used: 1558/8192, overflow=0
11.4055
    SARSA(λ=0.9) ...   IHT used: 1359/8192, overflow=0
7.6507
    Q(λ=0.9) ...   IHT used: 1864/8192, overflow=0
```

```
7.6507
    Actor-Critic ...   Critic IHT used: 539/8192, overflow=0
  Actor  IHT used: 1078/8192, overflow=0
7.6507

  -- asymmetric_rush --
    Q-learning ...   IHT used: 2694/8192, overflow=0
6.3928
    SARSA(λ=0.9) ...   IHT used: 1965/8192, overflow=0
6.3928
    Q(λ=0.9) ...   IHT used: 2892/8192, overflow=0
6.3928
    Actor-Critic ...   Critic IHT used: 1543/8192, overflow=0
  Actor  IHT used: 3086/8192, overflow=0
6.3928

Seed 42 complete -- checkpoint saved.

============================================================
SEED 7
============================================================

  -- symmetric_light --
    Q-learning ...   IHT used: 4408/8192, overflow=0
8.0600
    SARSA(λ=0.9) ...   IHT used: 4497/8192, overflow=0
6.3312
    Q(λ=0.9) ...   IHT used: 4584/8192, overflow=0
6.6045
    Actor-Critic ...   Critic IHT used: 2164/8192, overflow=0
  Actor  IHT used: 4328/8192, overflow=0
5.5917

  -- symmetric_heavy --
    Q-learning ...   IHT used: 1436/8192, overflow=0
11.1959
```

```
# Ensure evaluate_greedy_ac is defined (greedy policy for Actor-Critic)
# This mirrors evaluate_greedy but uses softmax argmax instead of Q argmax

def evaluate_greedy_ac(theta, iht_th, mode, arrival_rates,
                       eval_steps=10_000, warmup=2_000, seed=99):
    """Evaluate Actor-Critic greedy policy (take argmax of softmax probabiliti
    env = TrafficEnv(mode=mode, arrival_rates=arrival_rates, seed=seed)
    env.reset()
    total_r = 0.0
    for t in range(warmup + eval_steps):
        ts   = env.state_for_tiling()
        p_keep, p_switch = softmax_policy(theta, iht_th, ts)
        action = KEEP if p_keep >= p_switch else SWITCH
        _, r, _ = env.step(action)
        if t >= warmup:
            total_r += r
    return -total_r / eval_steps

print("evaluate_greedy_ac defined.")
```

```
evaluate_greedy_ac defined.
```

```
import numpy as np

ms = np.load("multi_seed_results.npy", allow_pickle=True).item()

METHODS = [
    ("ql",    "Q-learning (λ=0)"),
    ("sarsa", "SARSA (λ=0.9)"),
    ("qlam",  "Q(λ=0.9)"),
    ("ac",    "Actor-Critic"),
]
SCENARIOS_LIST = ["symmetric_light", "symmetric_heavy", "asymmetric_rush"]
FIXED = {s: baseline_results[s]["fixed"] for s in SCENARIOS_LIST}

print("\nMulti-seed results (mean ± std over seeds:", list(ms.keys()), ")")
print()
print(f"{'Method':<24}", end="")
for s in SCENARIOS_LIST:
    label = s.replace('_',' ').title()[:14]
    print(f"  {label:>18}", end="")
print()
print("-" * 80)

for key, label in METHODS:
    print(f"{label:<24}", end="")
    for scenario in SCENARIOS_LIST:
        vals = [ms[seed][scenario][key] for seed in ms]
        mean = np.mean(vals)
        std  = np.std(vals)
        print(f"  {mean:.2f} ± {std:.2f}  ", end="")
    print()

print()
print("Fixed timing (reference):", {s: f"{FIXED[s]:.2f}" for s in SCENARIOS_LI
print()

# Also print the LaTeX-ready version
print("\n-- LaTeX table rows (mean $\\pm$ std) --\n")
for key, label in METHODS:
    row = f"{label:<22}"
    for scenario in SCENARIOS_LIST:
        vals = [ms[seed][scenario][key] for seed in ms]
        mean = np.mean(vals)
        std  = np.std(vals)
        diff = mean - FIXED[scenario]
        sign = "$-$" if diff < 0 else "+"
        row += f" & {mean:.2f}$\\pm${std:.2f}"
    row += " \\\\"
    print(row)
```

```
Multi-seed results (mean ± std over seeds: [42, 7, 123] )

Method                        Symmetric Ligh      Symmetric Heav      Asymmetri
--------------------------------------------------------------------------------
Q-learning (λ=0)          8.07 ± 0.14    11.51 ± 0.31    7.58 ± 1.09
SARSA (λ=0.9)             6.51 ± 0.13    7.63 ± 0.03    6.39 ± 0.00
Q(λ=0.9)                  6.17 ± 0.32    7.63 ± 0.03    6.39 ± 0.00
Actor-Critic              5.87 ± 0.52    7.63 ± 0.03    6.39 ± 0.00
```

```
Fixed timing (reference): {'symmetric_light': '6.48', 'symmetric_heavy': '11.92


-- LaTeX table rows (mean $\pm$ std) --

Q-learning (λ=0)       & 8.07$\pm$0.14 & 11.51$\pm$0.31 & 7.58$\pm$1.09 \\
SARSA (λ=0.9)          & 6.51$\pm$0.13 & 7.63$\pm$0.03 & 6.39$\pm$0.00 \\
Q(λ=0.9)               & 6.17$\pm$0.32 & 7.63$\pm$0.03 & 6.39$\pm$0.00 \\
Actor-Critic           & 5.87$\pm$0.52 & 7.63$\pm$0.03 & 6.39$\pm$0.00 \\
```

19. Updated Table 1 with Error Bars